

Haptic Hand: Haplink Fingers and Delta Thumb

Holly Chen, Jared Scott, Kiran Marques, Rajiv Joshi,
Weichen Qiu, Yuichi Hirose

Abstract — This paper presents the design and implementation of a low-cost, 3D-printed haptic device that provides force feedback for hand interactions with virtual objects. The design consists of three mechanical subsystems that interface with a user's thumb and two fingers. Interactions between the fingers and objects of different sizes and stiffnesses can be visualized in a GUI. This research provides validation of an affordable, easily manufactured haptic device that is accessible and can be used for educational purposes.

I. INTRODUCTION

The human hand is capable of performing many complex tasks, but relies on feedback to execute them successfully. Kinesthetic feedback, which provides force information, is one method to improve human proprioception. The novel contribution of this research project is the design that renders force in multiple directions (unlike current systems that can only do closing or opening, not both). Applications for this project/haptic device are geared towards making haptics universally accessible for in-classroom interactive student learning at an affordable cost.

II. BACKGROUND

Designs that make use of force feedback to improve user interaction include grounded devices, which are mounted to a stationary surface, and ungrounded devices, which are directly mounted to the user's body. HIRO III is a grounded five-fingered haptic interface robot, which is placed opposite a hand, not on the hand [4]. The robot consists of a hand with five 3-DOF fingers, and an arm and wrist, each of which have 3-DOF. Hence, the entire system consists of a 15-DOF hand and a 6-DOF arm. Because users only need to put their fingertips into the finger holders, it is easy to wear and remove. Other grounded devices such as the Novint Falcon [3] and PHANTOM [7] sit on table tops and simulate hand-tool interactions with 3-DOF.

Ungrounded haptic devices include exoskeleton and glove designs that interface directly with the body. The Sensing and Force-Feedback Exoskeleton (SAFE) is a robotic glove that combines active and passive force feedback to each finger of a hand [6]. This system utilizes strain-gauges between the glove mechanism and fingertips to obtain accurate kinematics and forces. Wearable gloves are a good method for the situation where a user walks around wearing a VR headset. However, while they can represent force at the contact points, it is difficult to render the weight of virtual objects because the device is already weighing on the hand [4]. The specific wearable design such

as which part of the hand a device is grounded to also has a significant impact on the user's perception and experience, which itself is a domain of research [5].

While many designs exist to provide force feedback to users for virtual reality, medical training and rehabilitation purposes, these devices are often high cost which limits their accessibility to the general public. One area of research that remains to be explored is designing a haptic device that is affordable, open-source, and easy to manufacture for learning environments.

The DeltaZ robot, an accessible, compliant delta robot manipulator from the ZoomLab that is 3D printed at a low cost of \$50 USD [2]. A key advantage of this design is the replacement of typical ball joints with a single-piece 3D printed flexible joint from thermoplastic polyurethane (TPU) material and parallel 4-bar mechanisms [2]. This avoids issues like singularities and links breaking due to collisions that exist with traditional ball joints [2]. This DeltaZ mechanism formed the foundation of the thumb for this device.

III. GOAL

The overarching goal of this project falls into a subcomponent of a larger research project. The eventual goal is to develop a haptic interface device to control the three fingers (i.e. thumb, index, and middle fingers) of an existing foam robotic hand. In addition, the device should give haptic feedback back to the user.

Given the time constraints, the scope of this paper is limited to giving haptic feedback to three fingers to simulate a spherical virtual object and visualizing the interaction in a virtual environment.

The haptic device should be able to actuate the thumb in 3-DOF (circumduction, retropulsion, adduction, abduction) and the index and middle finger in 2-DOF (finger flexion, extension, bending) [1].

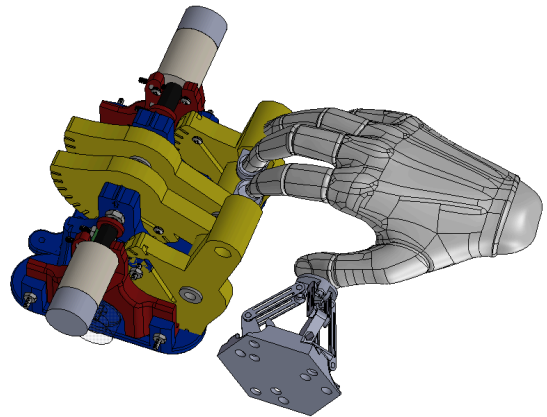


Figure 1: Design Concept. Credit to Prof. Orta Martinez

Haptic Hand: Haplink Fingers and Delta Thumb

Holly Chen, Jared Scott, Kiran Marques, Rajiv Joshi,
Weichen Qiu, Yuichi Hirose

Abstract — This paper presents the design and implementation of a low-cost, 3D-printed haptic device that provides force feedback for hand interactions with virtual objects. The design consists of three mechanical subsystems that interface with a user's thumb and two fingers. Interactions between the fingers and objects of different sizes and stiffnesses can be visualized in a GUI. This research provides validation of an affordable, easily manufactured haptic device that is accessible and can be used for educational purposes.

IV. INTRODUCTION

The human hand is capable of performing many complex tasks, but relies on feedback to execute them successfully. Kinesthetic feedback, which provides force information, is one method to improve human proprioception. The novel contribution of this research project is the design that renders force in multiple directions (unlike current systems that can only do closing or opening, not both). Applications for this project/haptic device are geared towards making haptics universally accessible for in-classroom interactive student learning at an affordable cost.

V. BACKGROUND

Designs that make use of force feedback to improve user interaction include grounded devices, which are mounted to a stationary surface, and ungrounded devices, which are directly mounted to the user's body. HIRO III is a grounded five-fingered haptic interface robot, which is placed opposite a hand, not on the hand [4]. The robot consists of a hand with five 3-DOF fingers, and an arm and wrist, each of which have 3-DOF. Hence, the entire system consists of a 15-DOF hand and a 6-DOF arm. Because users only need to put their fingertips into the finger holders, it is easy to wear and remove. Other grounded devices such as the Novint Falcon [3] and PHANToM [7] sit on table tops and simulate hand-tool interactions with 3-DOF.

Ungrounded haptic devices include exoskeleton and glove designs that interface directly with the body. The Sensing and Force-Feedback Exoskeleton (SAFE) is a robotic glove that combines active and passive force feedback to each finger of a hand [6]. This system utilizes strain-gauges between the glove mechanism and fingertips to obtain accurate kinematics and forces. Wearable gloves are a good method for the situation where a user walks around wearing a VR headset. However, while they can represent force at the contact points, it is difficult to render the weight of virtual objects because the device is already weighing on the hand [4]. The specific wearable design such as which part of the hand a device is grounded to also has a significant impact on the user's perception and experience, which itself is a domain of research [5].

While many designs exist to provide force feedback to users for virtual reality, medical training and rehabilitation purposes, these devices are often high cost which limits their accessibility to the general public. One area of research that remains to be explored is designing a haptic device that is affordable, open-source, and easy to manufacture for learning environments.

The DeltaZ robot, an accessible, compliant delta robot manipulator from the ZoomLab that is 3D printed at a low cost of \$50 USD [2]. A key advantage of this design is the replacement of typical ball joints with a single-piece 3D printed flexible joint from thermoplastic polyurethane (TPU) material and parallel 4-bar mechanisms [2]. This avoids issues like singularities and links breaking due to collisions that exist with traditional ball joints [2]. This DeltaZ mechanism formed the foundation of the thumb for this device.

VI. GOAL

The overarching goal of this project falls into a subcomponent of a larger research project. The eventual goal is to develop a haptic interface device to control the three fingers (i.e. thumb, index, and middle fingers) of an existing foam robotic hand. In addition, the device should give haptic feedback back to the user.

Given the time constraints, the scope of this paper is limited to giving haptic feedback to three fingers to simulate a spherical virtual object and visualizing the interaction in a virtual environment.

The haptic device should be able to actuate the thumb in 3-DOF (circumduction, retropulsion, adduction, abduction) and the index and middle finger in 2-DOF (finger flexion, extension, bending) [1].

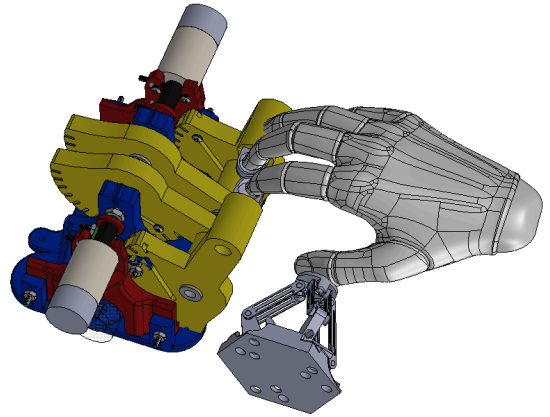


Figure 1: Design Concept. Credit to Prof. Orta Martinez

VII. METHODS

A. Mechanical Design - Index and Middle

When Evaluating which 2-DOF mechanism to implement for actuating both the index and middle finger the team established some baseline design metrics. 1) Both the index and middle finger mechanism should be the same

mechanical system, 2) The mechanism should be compatible with the 12V Pololu Motor - No Gearbox 25D mm, 3) The system should have decoupled motors, and 4) The overall mechanical system should be easy to manufacture in house at a minimal cost.

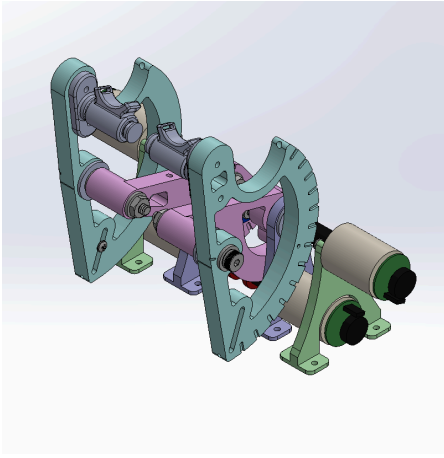


Figure 2: Finalized Fingers Mechanical Design

Now addressing these fundamental design criteria: While using distinct mechanical mechanisms for each finger might have improved the individual workspace of each finger, opting for identical mechanisms for the index and middle fingers simplified the overall device design. This approach minimizes the kinematic calculations and code implementation needed, as Finger Devices 1 and 2 shared the same set of equations and Jacobians – differing only by flipped encoder measurements. This will be analyzed and explained further in the following sections.

Our team's design was constrained to using 12V Pololu motors without gearboxes. While gearboxes in most mechanical systems typically enable higher torque outputs, they also introduce backlash due to the inherent spacing between gears. This backlash can degrade the quality of haptic feedback through ruining the user's ability to perceive fine tactile sensations. Therefore, we chose to avoid gearboxes. The selected motors had been successfully used in previous haptic devices and were proven effective for rendering educational haptic effects – such as virtual walls, surface textures, and shapes – using basic impedance controller implementations [8].

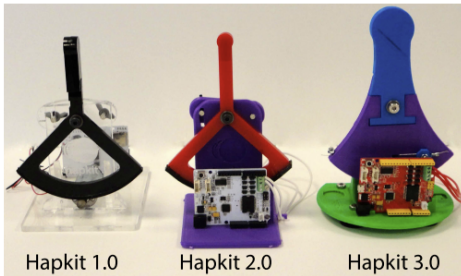


Figure 3: Various version of Hapkit Device that implements Capstan Drive with 12V Pololu Motor

These motors are equipped with quadrature encoders that offer a resolution of 48 counts per revolution (CPR). Due to

this relatively low resolution, our mechanical design had to compensate in order to achieve higher CPR resolution. This could be addressed by using more expensive motors, higher-resolution encoders, or carefully choosing mechanical transmission systems that increase the CPR relative to the end-effector rotation.

This need brought us to our third design requirement: selecting an appropriate mechanical transmission. We evaluated two potential options – a 2-bar linkage with belt drive and a capstan drive. The 2-bar linkage system is straightforward to manufacture and allows for simple forward kinematics to calculate end-effector position. However, a significant drawback is that the motors in such systems are inherently coupled, meaning the motion of one motor affects the motion of the other. Although this issue can be mitigated through torque feedback and closed-loop control, integrating torque sensors was not feasible within our limited project budget.



Figure 4: Phantom Omni Haptic Device

On the other hand, a capstan drive offers a key advantage: motor decoupling. Literature and experimental studies have shown that capstan mechanisms – essentially two cylinders acting as rolling bodies connected by a cable—allow partial decoupling of motor motion. This means the second linkage can rotate freely around the second motor shaft without directly affecting the encoder counts. Additionally, capstan drives are known for their smooth transmission characteristics, with minimal backlash—limited primarily to the friction between the cable and the capstan body. This smoothness makes capstan systems a popular choice in many haptic devices, such as the Phantom Omni [7].

Another major benefit of the capstan drive is the improved encoder resolution it provides. By using a larger capstan radius in contact with the motor's smaller output shaft, the effective CPR increases – since the motor must rotate more to achieve the same displacement at the end-effector compared to a direct drive system. This resolution enhancement becomes especially important in mechanisms like the delta-style thumb, which we will discuss later.

As a result, a dual-capstan drive was selected as the optimal mechanical transmission for our device. It offered a cost-effective, smooth, and compact solution for torque transfer, ideal for a lightweight and space-efficient haptic system.

B. Evaluation of Finger and Device Workspace

The first task in designing the finger-interaction portion of the Haptic Hand was to determine the effective linkage lengths of the device. Independent of the exact method used to design these links, for example rigid links or capstan drive, it was decided that there would be two links for two degrees of freedom. The finger workspace shown in blue in Figure 5 was plotted referring to an individual with a longer middle finger length. A long middle finger was chosen to create the largest workspace that the device would need to operate in so that it would be accessible to the majority of users consisting of long and short finger lengths.

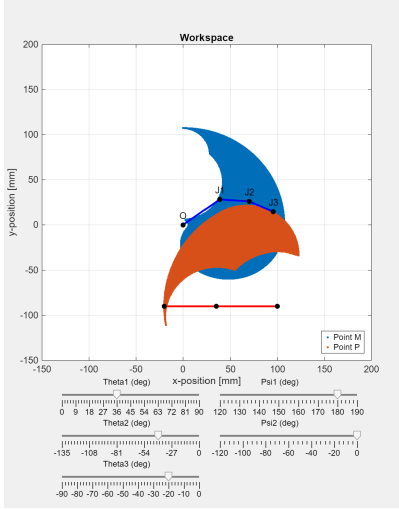


Figure 5: Device workspace (orange) overlaid on finger workspace (blue)

The device workspace shown in orange was plotted over the finger workspace with varying linkage lengths until the most desirable workspace was experimentally achieved. Desirable was defined to be sufficient overlap such that the finger could experience the largest range of motion in the natural “belly” region of the finger workspace while also having the device linkage lengths be as small as possible for device compactness and weight savings. The final ideal linkage lengths were determined to be $L1 = 65$ mm, $L2 = 55$ mm as shown in Figure 6.

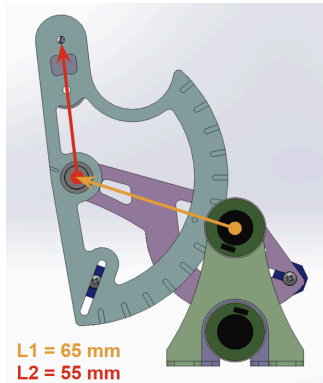


Figure 6: Optimized Fingers Device Linkage Lengths

C. Software Design (Fingers)

C.1. Individual Finger Software

Finger 2 matched the kinematics of the class Haplink device shown in Figure 7. The top-down view of the Haplink device became a right-facing view of the Haptic Hand device.

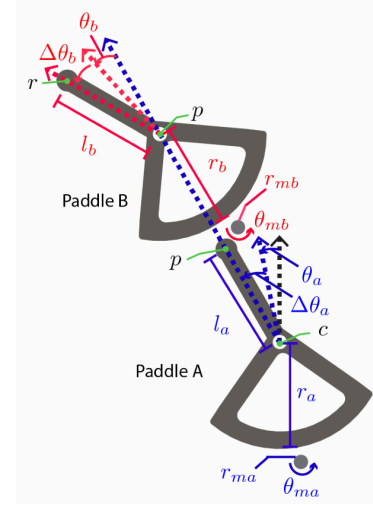


Figure 7: Kinematics of Haplink and Finger 2 of Haptic Hand

This enabled the team to use the same kinematic equations and Jacobian for Finger 2, only changing the physical parameters for $\Delta\theta_a$ and $\Delta\theta_b$, which are the offsets of the handles relative to the center line, along with l_a , l_b , r_a , and r_b . These equations are shown below in Figure 5.

$$\begin{aligned} \begin{bmatrix} p_x \\ p_y \end{bmatrix} &= \begin{bmatrix} -l_a \sin(\tilde{\theta}_a) + c_x \\ l_a \cos(\tilde{\theta}_a) + c_y \end{bmatrix} & \begin{aligned} \tilde{\theta}_a &= \theta_a + \Delta\theta_a \\ \tilde{\theta}_b &= \theta_b + \Delta\theta_b \end{aligned} \\ \begin{bmatrix} r_x \\ r_y \end{bmatrix} &= \begin{bmatrix} -l_b \sin(\tilde{\theta}_a + \tilde{\theta}_b) + p_x \\ l_b \cos(\tilde{\theta}_a + \tilde{\theta}_b) + p_y \end{bmatrix} & \begin{aligned} \theta_a &= -\frac{r_{ma}}{r_a} \theta_{ma} \\ \theta_b &= -\frac{r_{mb}}{r_b} \theta_{mb} - \frac{r_{mb}}{r_b} \frac{r_{ma}}{r_a} \theta_{ma} \end{aligned} \end{aligned}$$

$$J = \begin{bmatrix} \frac{\partial r_x}{\partial \tilde{\theta}_a} & \frac{\partial r_x}{\partial \tilde{\theta}_b} \\ \frac{\partial r_y}{\partial \tilde{\theta}_a} & \frac{\partial r_y}{\partial \tilde{\theta}_b} \end{bmatrix} = \begin{bmatrix} -l_b \cos(\tilde{\theta}_a + \tilde{\theta}_b) + -l_a \cos(\tilde{\theta}_a) & -l_b \cos(\tilde{\theta}_a + \tilde{\theta}_b) \\ -l_b \sin(\tilde{\theta}_a + \tilde{\theta}_b) + -l_a \sin(\tilde{\theta}_a) & -l_b \sin(\tilde{\theta}_a + \tilde{\theta}_b) \end{bmatrix}$$

Figure 8: Kinematic equations and Jacobian for Finger 2

Physically, Finger 1 is simply Finger 2 mirrored across the right-facing plane between them. From a kinematic perspective, however, it was useful to think of Finger 1 not as kinematically mirrored, but as kinematically identical with the sole change being the default direction of the motors. This is how the fingers act from a right-facing perspective, shown in Figure 9.

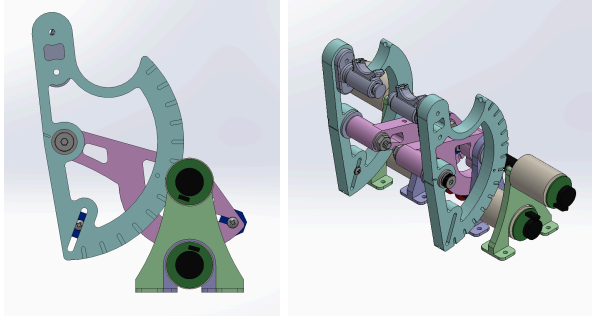


Figure 9: Kinematically identical fingers from right-facing view (left) and isometric view (right)

In the base Haplink code as well as Finger 2, motor encoders A and B increment positively in the clockwise direction and negatively in the counterclockwise direction relative to the right-facing view shown in Figures 4 and Figure 6 (left), which is the opposite of the reference directions shown for θ_{ma} and θ_{mb} in Figure 4. To perform calculations in accordance with the reference directions specified in Figure 7, the base Haplink code therefore included a negative sign after retrieving the motor encoder position so that during calculations, it was as if motor encoders A and B incremented positively in the counterclockwise direction and negatively in the clockwise direction in alignment with the reference directions. After calculating the required torques to provide the motors with according to this reference, a negative sign was applied to these torques so that they once again matched the hardware direction of the encoders and rendered properly for the user. This procedure of transforming from hardware space to reference space, doing calculations, then transforming back from reference space to hardware space was already implemented in the base Haplink code and is visualized in Figure 10.

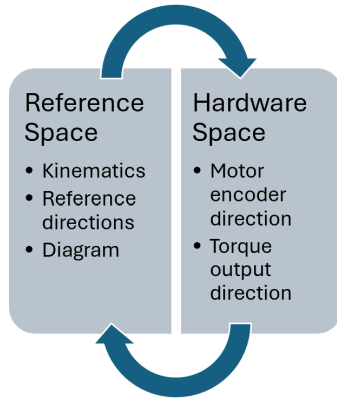


Figure 10: Relationship between reference space and hardware space

When Finger 1 was mirrored, the positive encoder direction of the motors became aligned with the positive counterclockwise direction in the Figure 4 diagram. This meant that the sole difference between code for Fingers 1 and 2 were the negative signs applied after retrieving motor position and before setting output torque for both motors. In

other words, no transformation between reference space and hardware space was necessary for Finger 1 as they were identical. Making this sign change in the Finger 1 functions enabled identical finger kinematics and preserved axes alignment between the two fingers.

C.2. Finger Integration

The fingers were integrated with the delta code so that all could respond to feeling the same virtual object. This was tested with a sphere. The global coordinate system was placed 37mm off the baseplate of the thumb delta at the centerpoint of the three motors. The coordinate systems of the delta and fingers were aligned as shown in Figure 11.

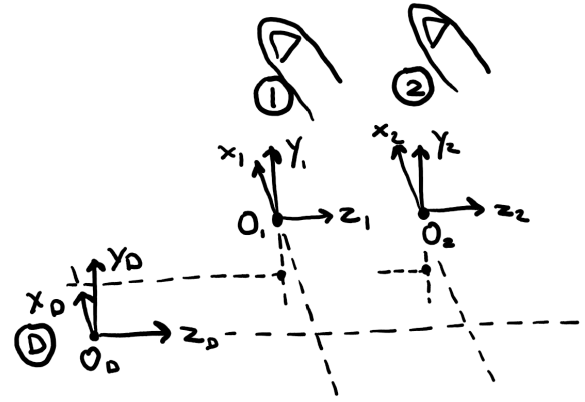


Figure 11: Delta, Finger 1, and Finger 2 coordinate systems

Aligning all coordinate systems enabled transformations from finger coordinates to global coordinates to consist solely of translations. These utilized the normal x , y , and z distances from the global origin to each of the finger origins.

The steps for rendering a virtual object defined in the global coordinate system for each of the fingers was as follows:

1. Map the position of the finger end effector coordinates into the global coordinate systems via translations.
2. Check if the finger end effectors are within the virtual sphere.
3. Calculate the force vector pushing each finger out of the sphere:

$$F_x = K_{sphere} * (R_{sphere} - P_{user-sphere}) * n_x \text{ where}$$

F_x is the force in the x direction, K is the spring stiffness of the sphere, R is the radius of the sphere, P is the normal distance from the finger end effector to the center of the sphere, and n_x is the x component of the unit vector pointing from the center of the sphere to the finger end effector. This calculation is further broken down in the supplied project code.

4. Map the force vectors back to their respective local coordinate systems and ignore the z component, as fingers are 2-DOF. Mapping in this step required no action as all axes (global, Finger 1, Finger 2) were aligned.

5. Map the force components to motor torques for each finger using the Jacobian to render force to the user, making sure to apply the sign changes described at the end of section B.1. The Jacobian was the same for both fingers.

D. Mechanical Design - Thumb

The thumb section was designed to provide 3 degrees of freedom (DOF), as unlike the index and middle fingers, which mainly perform flexion and extension movements, the thumb also frequently performs abduction and adduction. To achieve 3 DOF, a delta mechanism was adopted. An alternative considered was to mount a 2 DoF device on a rotating table, like the Phantom Omni (Figure). However, in that configuration, the smoothness of each joint was expected to significantly affect the haptic feedback. Given the limited time before the final presentation, there were concerns about whether parts made with a 3D printer or laser cutter could achieve the required smoothness, which led to the decision to use a delta mechanism.

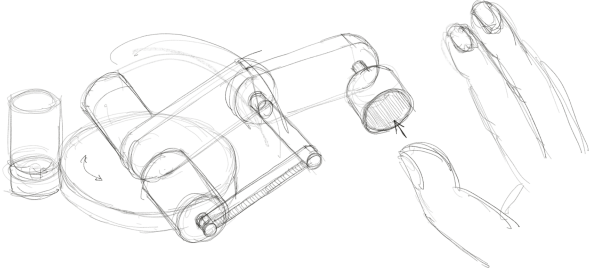


Figure 12: Alternate Thumb Device Design

For the delta mechanism, parts from the Delta Z [2] were used. These parts utilize TPU filament and implement a compliant mechanism that enables the delta mechanism's joints to be realized as a monolithic 3D print.

To achieve the haptic sensation where users feel being pushed back when they are touching a virtual sphere, we need to calculate the torques of each motor. To do this, we used the inverse kinematics using a Jacobian matrix ($F = J^T T$, where F is force and T is torque).

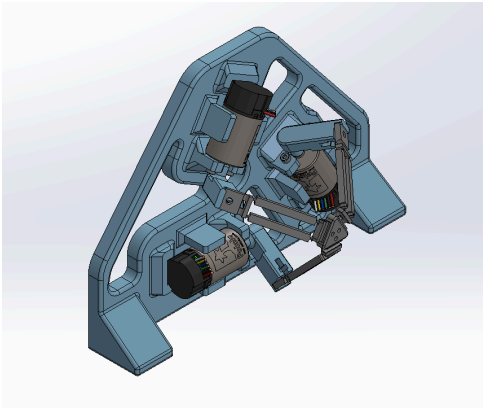


Figure 13: Alternate Thumb Device Design

E. Hand Visualization (Processing)

The Processing programming language was used to render a visual representation of the user's interaction with the device. A diagram of this process is shown in Figure 14.

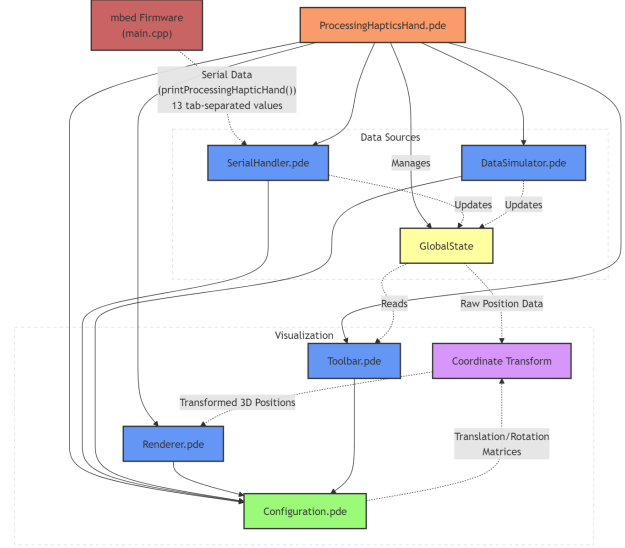


Figure 14: Haptic hand visualization data pipeline

The mBed C/C++ code controlling the integrated device is configured to repeatedly output a string containing 13 values: the x-, y-, and z-values of the 3 finger positions (9 values total), the x-, y-, and z-values of rendered sphere centroid, and the radius of the rendered sphere, all in the global reference frame. These values are parsed by a SerialHandler.pde Processing function, and translations and rotation matrices are applied to ensure that these values display correctly in a 3D graphics environment. Renderer.pde filters and displays the transformed values from mBed in real time for the user to visualize their interactions with the virtual object, as shown in Figure 15.

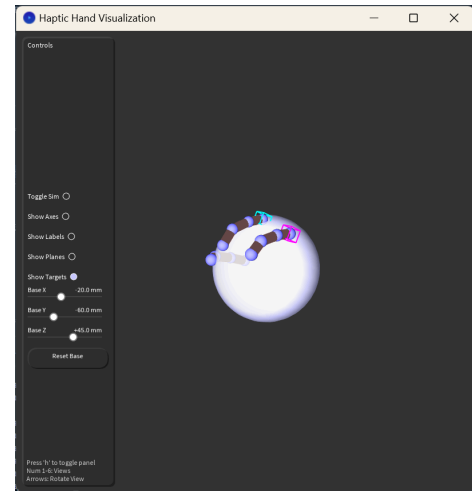


Figure 15: Haptic hand visualization program

VIII. EVALUATION AND RESULTS

The haptic feedback of the thumb is found unstable. The cause we expect behind this is the low resolution of the encoder attached to the motor. The encoder has 12 per revolution resolution, which means 48 per revolution with the quadrature. In this case, the smallest change in angles the encoder detects is 7.5° ($360^\circ / 48$), which apparently affects the calculation of the end effector from the motor angles in the forward kinematics as well as the calculation of the torques from the desired force to apply in the inverse kinematics. The most straightforward solution is to use encoders with higher resolution. Another option would be reduction using gears or capstans. Currently, the arms of the delta are directly connected to the motor shafts, so the low resolution of the encoder affects the entire system as is. With the reduction, we can increase the resolution of the entire system, although we need to pay attention to changes in the motor torques because of this.

IX. CONCLUSIONS AND FUTURE WORK

This project was a proof of concept design for a low-cost haptics device that renders virtual objects. The scope was a small section of a larger research project so many iterations can still be made to improve the system. The thumb delta mechanism has low resolution due to direct drive of the motors. To improve the smoothness of the force feedback, the motor resolution needs to be amplified. This can either be accomplished by using nicer motors or adding a capstan drive type of mechanism that scales the resolution of the motors. For the finger system, the capstan handle lengths can be optimized to increase the workspace and allow for larger virtual object interactions. Additionally, the friction and weight of the capstans can be felt by the user. The weight that is felt can be reduced by designing a counter-weight or adjusting the design to have a more symmetric center of mass. For software, improvements can be made to the mapping of delta movement and thumb visualization. Currently, the physical delta coordinates are not 1-to-1 with the visualization so a rotation will need to be implemented to transform the coordinate systems to match.

APPENDIX

A: Delta Kinematics Derivation (see attached document at end of paper)

ACKNOWLEDGMENT

The authors would like to acknowledge the advice and guidance provided by Professor Melisa Orta Martinez. Funding for this project was provided by the SHRED lab at Carnegie Mellon University. This project was completed as the final project in 16880-A: Engineering Haptic Interfaces class in Spring 2025.

REFERENCES

- [1] Edison-Metuchen Orthopaedic Group. "Normal Hand Anatomy". <https://www.emog.net/normal-hand-anatomy/>.
- [2] S. Patil, S. C. Alvares, P. Mannam, O. Kroemer and F. Z. Temel, "DeltaZ: An Accessible Compliant Delta Robot Manipulator for Research and Education," 2022 IEEE/RSJ International Conference

- on Intelligent Robots and Systems (IROS), Kyoto, Japan, 2022, pp. 13213-13219, doi: 10.1109/IROS47612.2022.9981257.
- [3] Sirwal, S. A., Ahmad, B., & Koul, M. H. (2024). Haptic Rendering of Viscous Media. 8th Conference on Advances in Control and Optimization of Dynamical Systems ACODS 2024, 57, 226–231. <https://doi.org/10.1016/j.ifacol.2024.05.039>
- [4] Endo T, Kawasaki H, Mouri T, Ishigure Y, Shimomura H, Matsumura M, Koketsu K. Five-Fingered Haptic Interface Robot: HIRO III. IEEE Trans Haptics. 2011 Jan-Feb;4(1):14-27. <https://doi.org/10.1109/WHC.2009.4810812>.
- [5] S. Nisar, M. O. Martinez, T. Endo, F. Matsuno and A. M. Okamura, "Effects of Different Hand-Grounding Locations on Haptic Performance With a Wearable Kinesthetic Haptic Device," in IEEE Robotics and Automation Letters, vol. 4, no. 2, pp. 351-358, April 2019, <https://doi.org/10.1109/LRA.2018.2890198>.
- [6] Ben-Tzvi, Pinhas, and Zhou Ma. "Sensing and Force-Feedback Exoskeleton (SAFE) Robotic Glove." IEEE Transactions on Neural Systems and Rehabilitation Engineering 23, no. 6 (November 2015): 992–1002. <https://doi.org/10.1109/TNSRE.2014.2378171>.
- [7] J. K. Salisbury and M. A. Srinivasan, "Phantom-based haptic interaction with virtual objects," in IEEE Computer Graphics and Applications, vol. 17, no. 5, pp. 6-10, Sept.-Oct. 1997, doi: 10.1109/MCG.1997.1626171.
- [8] M. Orta Martinez, C. M. Nunez, T. Liao, T. K. Morimoto and A. M. Okamura, "Evolution and Analysis of Hapkit: An Open-Source Haptic Device for Educational Applications," in IEEE Transactions on Haptics, vol. 13, no. 2, pp. 354-367, 1 April-June 2020, doi: 10.1109/TOH.2019.2948609.